

ECE 657A: Effective Deep Learning Using Backlinks to Make DL More Robust

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Types of Deep Learning

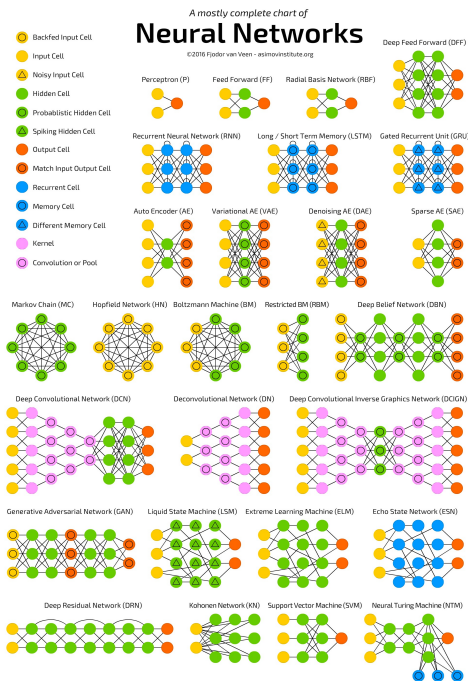
Mostly complete chart of Neural Networks:

- **asimovinstitute.org**

Probably out of date already, but a nice way to think about them, look for new architectures.

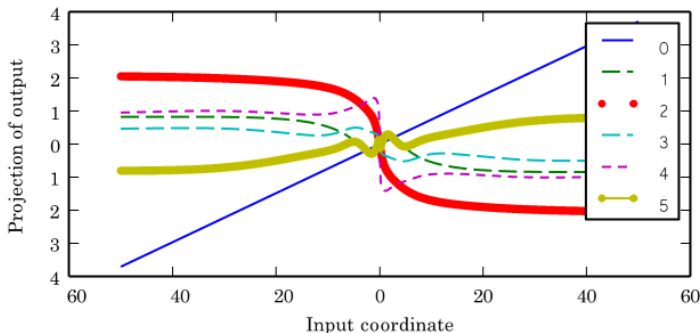
Note: some older algorithms can be represented this way

- - SVM
- - RBF
- - Perceptron
- - Markov Chain



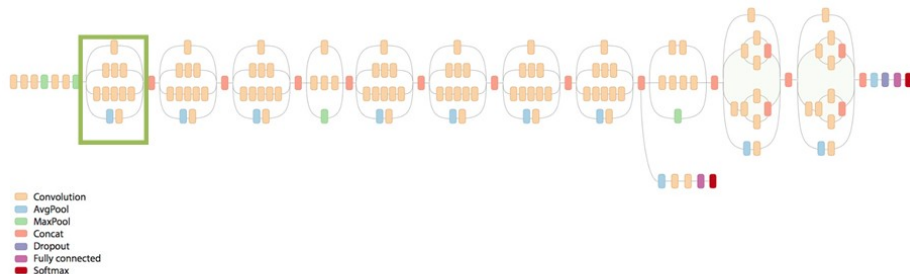
The Exploding/Vanishing Gradient Problem

- Composing many nonlinear functions (very deep or across time) what happens?
- As the gradient is back-propagated to earlier layers, repeated multiplication may makes the gradient more and more extreme and result is nonlinear and erratic.



Inception Networks

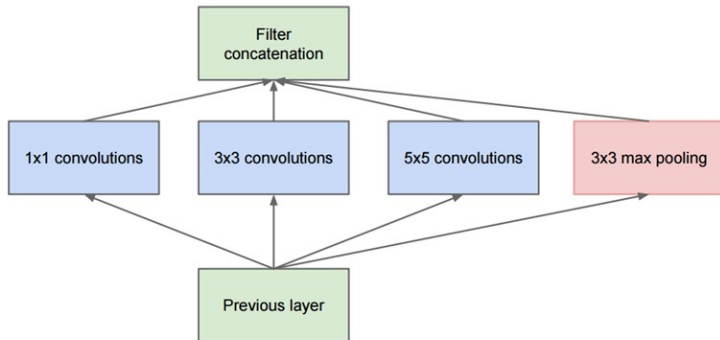
More complex architecture designed to achieve high accuracy with low computational cost introduced in **GoogLeNet** (Szegedy, 2015)



Green box shows parallel region of GoogLeNet

Inception Modules

A basic inception module:

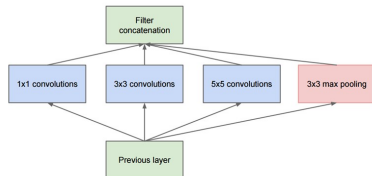


Naïve idea of an Inception module

- The data flows from **bottom** to **top**.
- The filter concatenation simply layers the outputs side by side.

Inception : Sure...but why?

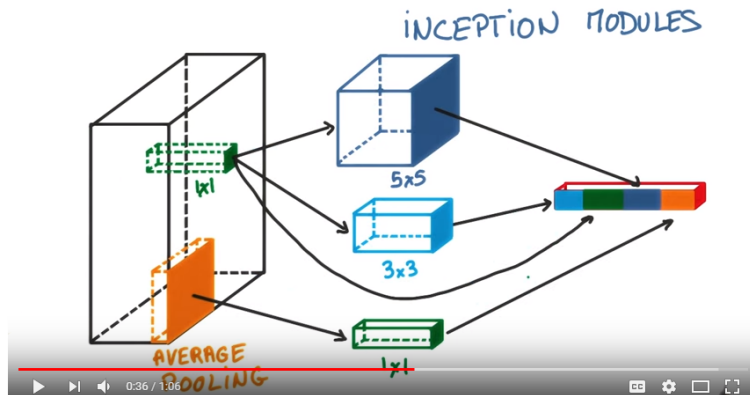
- Reduces the number of parameters to fit
- faster to train and test
- **OR!** same train/test speed on a BIGGER model?



Naive idea of an Inception module

- *In other words*, it allows you to go much deeper, more independent cuts through the data.
- In practice does not do much worse, or even does better than dense models..
- But what about 1x1?
 - “projection operation”, “projection layer”, “channel pooling layer”
- Read more :
 - <https://iamaaditya.github.io/2016/03/one-by-one-convolution/>
 - <https://machinelearningmastery.com/introduction-to-1x1-convolutions-to-reduce-the-complexity-of-convolutional-neural-networks/>

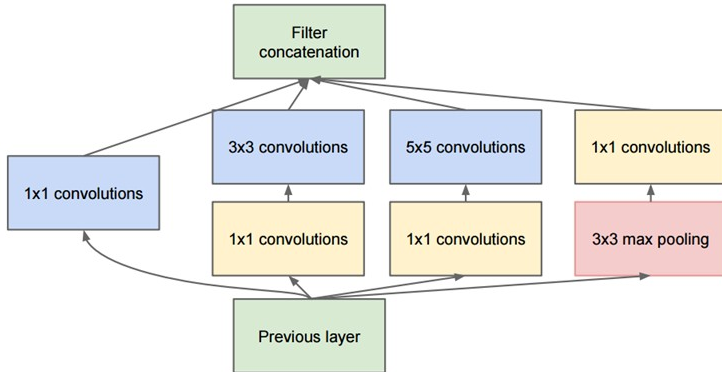
Filter Concatenation



- Good description
 - <https://www.youtube.com/watch?v=VxhSouuSZDY>

Inception Modules

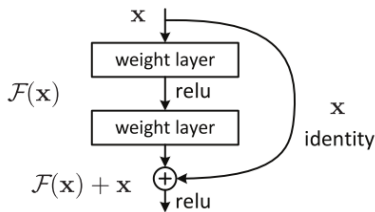
A larger inception module, using 1x1 layers to reduce data first:



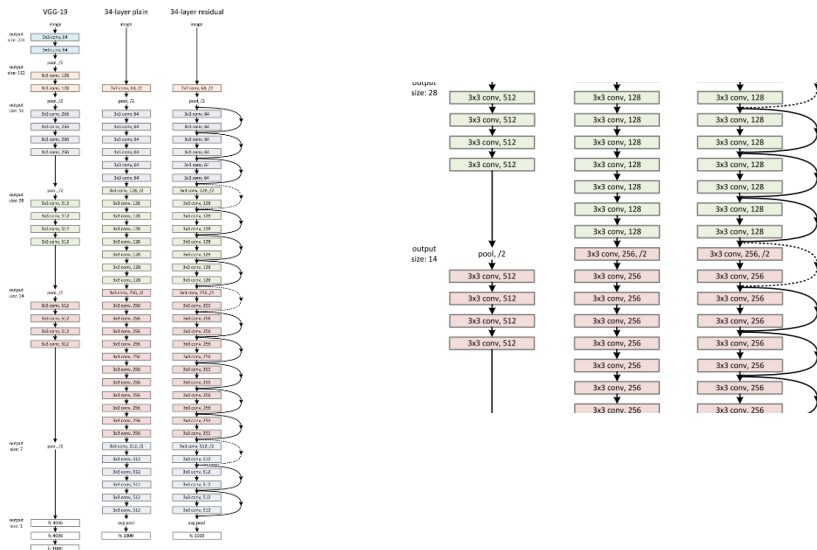
Full Inception module

Residual Networks

- ResNet: blocks of layers with same spatial size, with each layers output added to the same buffer that is repeatedly updated.
- Very many updates = very deep net (over 150 layers!)
- but without **vanishing gradient**.



Residual Networks



Examples